

COURSE TITLE : ANALOG CIRCUITS LAB
COURSE CODE : 3219
COURSE CATEGORY : B
PERIODS/WEEK : 5
PERIODS/SEMESTER: 75
CREDITS : 3

On completion of this course the student will be able to

1. To design and construct.
 - (i) RC differentiator circuit.
 - (ii) RC integrator circuit and study its pulse response.
2. To design, construct and test shunt diode clipper circuits.
 - (i) Positive clipper.
 - (ii) Negative clipper.
 - (iii) Biased clipper.
3. To construct a Zener diode clipper circuit and observe its input /output wave forms.
4. To design, construct and test various diode clamping circuits.
 - (i) Positive clamper.
 - (ii) Negative clamper.
 - (iii) Biased clamper.
5. To design single stage RC coupled CE amplifier for a given gain.
 - (i) Observe the phase difference between input and output wave forms.
 - (ii) Measure mid band gain.
 - (iii) Plot its frequency response and determine the band width.
6. To design an emitter follower circuit and measure the gain.
7. To design a RC phase shift oscillator for a given frequency of oscillation.
8. To design a simple sweep circuit and observe its output wave form for a square wave input.
9. To design a BJT stable multivibrator and plot the wave forms at base and collector of the transistors.
10. To determine the parameters of operational amplifier.
11. To design and set up inverting and non-inverting OP amp.
12. To design and set up adder and subtractor circuits using OP amp.
13. To design and set up Schmitt trigger circuit using OP amp.
14. To design and set up integrator and differentiator circuit using OP amp.
15. To design and set up current to voltage converter using OP amp.
16. To design and set up voltage to current converter using OP amp..
17. To design and set up an instrumentation amplifier and find the CMRR.